Contact Velocity Odometry - Structural UML Overview

OroGen Estimator Component

«input port» pose_imu
base::samples::RigidBodyState

«input port» joint_samples
base::samples::Joints

«input port» measurement_imu
base::samples::IMUSensors

«external» TaskBase

«implemented» contact_velocity_odometry_estimator::Task

-odometry : contact_velocity_odometry::Odometry
-initial_state : RobotState
-noise_params : NoiseParams
-measurement_noise : double
-chi_squared_threshold : double
-contact_md2_threshold : double
-estimate_contacts : bool
-wheel_names_speed_map : vector<pair<string,float>>
-contact_frame_transforms : vector<Affine3d>
-wheel_frame_fixed_transforms : vector<Affine3d>
-feet_to_fixed_wheel_frames_transforms : vector<Affine3d>

+configureHook() : bool
+updateHook() : void
+measureIMU() : void
+measureGyroBias() : void
+setInitialGyroBias() : void
+propagateIMU() : void
+correctContactVelocity() : void
+writeOutput() : void
+constructKinematics(ts, joints) : int

«output port» pose_body_estimated
base::samples::RigidBodyState

«output port» pose_body_estimated_local
base::samples::RigidBodyState

«external» RTT / Orocos Ports

«external» Transformer

«external» odometry::BodyContactState

«external» Eigen

«external» KDL

«external» URDF Model

input
input
input
reads/writes ports
gets frame transforms
output
output
reads robot configuration
owns

Contact Velocity Odometry Library

«implemented» contact_velocity_odometry::Odometry

-filter : unique_ptr<inekf::InEKF>
-robot_config : RobotConfiguration

+initialize(initial_state, noise_params) : void
+initialize(initial_state) : void
+setState(state) : void
+getState() : RobotState
+getStateCovariance() : MatrixXd
+getKalmanGain() : MatrixXd
+getInnovations() : MatrixXd
+setNoiseParams(params) : void
+setEstimateBias(estimate_bias) : void
+setRobotConfiguration(config) : void
+propagate(imu_measurement_vector, dt) : void

+correctContactVelocity(...) : void
+computeVectorKinematics(...) : void
+computeExpectedContactVelocity(...) : void
+computeMeasuredPotentialContactVelocity(...) : void

writes LF/MD contacts
matrices/vectors
contact velocity twist
parses wheel axes
converts state
builds PC/wheel kinematics
reads noise properties
creates initial state
robot model info
owns filter
wrapper state
wrapper noise

«data» contact_velocity_odometry::RobotConfiguration

+number_of_wheels : size_t
+number_of_contacts_per_wheel : size_t
+robot_urdf_path : string

«data» contact_velocity_odometry::RobotState

+rotation : Matrix3d
+velocity : Vector3d
+position : Vector3d
+gyroscope_bias : Vector3d
+accelerometer_bias : Vector3d

«data» contact_velocity_odometry::NoiseParams

+noise_gyroscope : double
+noise_accelerometer : double
+noise_gyroscope_bias : double
+noise_accelerometer_bias : double

InEKF Library

«external» inekf::InEKF

-error_type_ : ErrorType
-estimate_bias_ : bool
-state_ : RobotState
-noise_params_ : NoiseParams
-kalman_gain_ : MatrixXd
-innovations_ : MatrixXd
-contacts_ : map<int,bool>

+InEKF(state, params)
+getState() : RobotState
+setState(state) : void
+setNoiseParams(params) : void
+setEstimateBias(estimate_bias) : void
+Propagate(imu, dt) : void
+CorrectContactVelocity(residual, jacobian, noise) : void
+getKalmanGain() : MatrixXd
+getInnovation() : MatrixXd

«external» inekf::Kinematics

+id : int
+pose : Matrix4d
+covariance : Matrix<double,6,6>

«external» inekf::RobotState

-X_ : MatrixXd
-Theta_ : VectorXd
-P_ : MatrixXd

+getRotation() : Matrix3d
+getVelocity() : Vector3d
+getPosition() : Vector3d
+getGyroscopeBias() : Vector3d
+getAccelerometerBias() : Vector3d
+getP() : MatrixXd
+setP(P) : void

«external» inekf::NoiseParams

converts noise
filter noise
filter state

Stereotypes
«implemented» Implemented for thesis
«external» External dependency
«data» Data / configuration structure
«input port» OroGen input port
«output port» OroGen output port